

Husein Sidharta Muhammad

AI/ML ENGINEER

Dusun Besalen, RT. 06, No. Rumah 47, Kel. Baturetno, Kec. Banguntapan, Kab. Bantul, DI Yogyakarta 55197
huseinmuhammad@student.ub.ac.id • +62 819 0432 7982

EDUCATION

- Brawijaya University**
Bachelor of Information Technology
Awardee of Education Scholarship of Indonesia (BPI) 2024 (August 2024 – Ongoing)
• GPA: 3.8/4

Malang, Indonesia
August 2024 – Ongoing
- National University of Singapore – Summer Program**
Two-week hands-on course: AI for Community Impact

Singapore
19 June 2023 – 1 July 2023
- Islamic State Senior High School (MAN) 2 Yogyakarta**
The Most Outstanding in the Class – Academic Category
Awardee of Indonesia Advanced Country Scholarship (BIM) Study Abroad Preparation Program
• GPA: 91.67/100 • SAT: 1220

Yogyakarta, Indonesia
June 2021 – July 2024

LEADERSHIP AND PROJECTS

- SIGNESIA — BISINDO (Indonesian Sign Language) Cultural Preservation App**
Malang, Indonesia · Jul 2026 – Ongoing
Brawijaya University — NLP Engineer (Text-to-Gloss)
Developing the NLP pipeline for SIGNESIA, a mobile app that preserves Indonesian Sign Language (BISINDO) by scanning QR artifacts, selecting a regional dialect (Malang, Surabaya, or Jakarta), and playing narration with per-word BISINDO skeleton animation. Designated to develop an Indonesian-to-gloss (Text-to-Gloss) translator.
- BRONE — Indonesian Brawijaya University Voice Assistant Robot**
Malang, Indonesia · Aug 2026 – Ongoing
Brawijaya University — Speech-to-Text (ASR) & LLM Engineer
Built the speech-to-text and LLM subsystems for BRONE, an Indonesian voice assistant (STT → LLM → TTS) targeting under 2.5-second time-to-first-speech. For ASR, benchmarked Faster Whisper tiny/base/small on the Intel NUC and cut live word-error-rate from 17.1% to 7.2% via real-time noise suppression (RNNoise) and domain-aware handling of BRONE, FILKOM, and Brawijaya terms. For the LLM, tackled the ~26-second response latency of the GPT-4o baseline: experimented with a local Qwen 3.5 0.8B model, which was fast but lacked reasoning quality, then moved to DeepSeek V4 Flash as a smart, cheap, and fast alternative.
- Collaborative Research — Calibrated Safety Layer for Preventing Expert Mis-Dispatch in Radiology Vision-Language Model VILA-M3**
Malang, Indonesia · Aug 2026 – Ongoing
Intelligent System Laboratory & Kalbe Digital Lab — Member
Co-authored a GEMASTIK XIX scientific paper and qualify to the final, proposing a calibrated safety layer (a confidence-gated router) built on BioMedCLIP + MLP that prevents expert mis-dispatch in the radiology vision-language model VILA-M3. The router reduced the baseline expert mis-dispatch rate from 8% to 0% on 200 evaluation samples (McNemar's test, $p < 0.001$), outperformed the best parser strategy by +17.5 percentage points, and achieved routing consistency of 1.000 across 28,600 image–query pairs.
- Intelligent System Laboratory – Computer Science Faculty, Brawijaya University**
Malang, Indonesia · Aug 2025 – Ongoing
Operasional Division
To actively do research and collaboration in newest AI topics with other lab members as well as laboratory lecturers. By joining the lab, I get access to resources such as NVIDIA's high-end GPUs. As one of the members of the operational division, besides doing research, I also help the laboratory document the lab's events and post them to social media.
- English Study Students Forum (FORMASI), Brawijaya University**
Vice Coordinator Narrative Arts Division
FORMASI is a student activity unit providing opportunities Brawijaya University Students to boost their English skills. I actively contribute to the regular practice by preparing power point materials and reaching out a coach for the practice.

Malang, Indonesia
September 2025 – Ongoing

UABT is a student activity unit facilitates Brawijaya University Students to play badminton regularly. I contribute to the division by proposing external media partnership. I am also the person in charge for UABT anniversary, arranging event rundown, designating place of the event, and developing the staff executor structure.

SKILLS & INTERESTS

- **Technical:** LLM engineering, Python programming basic and advanced (Object-Oriented Programming), Data Science and Machine Learning, Computer Vision, Full Stack Web Development.
- **Libraries:** PyTorch, TensorFlow, NumPy, pandas, Matplotlib, scikit-learn.
- **Language:** B2 in English (IELTS 6), native proficiency in Bahasa Indonesia.
- **Laboratory:** Prominent skills and experience in varied research methods.
- **Interests:** Artificial Intelligence and scientific research projects.

AWARDS AND ACHIEVEMENTS

- Finalist of Scientific Paper GEMASTIK 2026 (Topic in Visual Language Model Radiology)
- 3rd Place of National IEEE Fest UB Research Plan PKM-KC 2025 (Topic in Machine Learning)
- Favorite Champion of Student Creativity Program Writing Competition (LP2PKM) Brawijaya University 2025 – Creative Initiative (KC) Category at the University Level
- Talented Student of Student Creativity Program Writing Competition (LP2PKM) Brawijaya University 2025 – Creative Initiative (KC) Category at the University Level
- The Best Intern Staff at University Students’ Research Group (K-RISMA) Brawijaya University Internship Program 2024
- 3rd Place of English Debate Competition of FORCE 2024 at the University Level
- 1st Place of Dekan Cup Computer Science Faculty Brawijaya University 2024 (Research Competition) – Information Technology Field at the Faculty Level
- Maximum Predicate (A-Distinction) in the Summer Program “AI for Community Impact” at the National University of Singapore 2023
- 1st Place in National Young Scientific Research Competition “Digital Literacy Strategy to Prevent Cybercrime and Hoax” 2022 by Surabaya State University
- Finalist of the National Level Young Researcher Competition (LPB) in the Field of Economics 2022 by the Center for Young Scientists
- 2nd Place in the Province Level Young Researcher Competition (LPB) in the Field of Economics 2022 by the Center for Young Scientists
- Top 30 National Level Scientific Research Competition “Young Generation Innovation in Enhancing Indonesia’s Economics” 2022 by Management Study Program UIN Sunan Ampel Surabaya

CERTIFICATION

Dasar Linux Untuk Cyber Security	Cyberkarta · 2026
Python for Data Science and Machine Learning Bootcamp	Udemy · 2025
NLP - Natural Language Processing with Python	Udemy · 2025